Hariharan Jayachandran

Chennai, India | +91 9342970272 | hariharanjayachandran04@gmail.com | linkedin.com/in/hariharan2604 | github.com/hariharan2604 | hariharanjayachandran.vercel.app

PROFESSIONAL SUMMARY

Java Backend Engineer with hands-on experience building and maintaining enterprise-grade microservices and RESTful APIs for large-scale e-commerce systems at Tata Consultancy Services. Strong proficiency in Java, Spring Boot, Spring MVC, Spring Data JPA/Hibernate, Spring Security, SQL, and DB2/MySQL. Proven experience improving API performance and build efficiency, with a strong understanding of backend architecture, database optimization, and Agile software development. Experienced with Docker, AWS, Git, and API testing.

EXPERIENCE

Tata Consultancy Services (TCS) – Systems Engineer Aug 2025 – Aug 2026

Chennai, India

- Design and develop enterprise backend systems using Java and Spring Boot, contributing to 4–6 microservices and RESTful APIs for a large-scale e-commerce client.
- Build and maintain high-throughput REST APIs supporting distributed e-commerce workflows, focusing on performance, reliability, scalability, and clean service interfaces.
- Implement data persistence using Spring Data JPA/Hibernate and DB2, including SQL query design and optimization for backend operations.
- Apply Spring Security for authentication and access control across backend services.
- Collaborate with QA, Business Analysts, and Product Owners in Agile ceremonies including Sprint Planning, Stand-ups, and Retrospectives to deliver production-ready features.
- Use Git and established software engineering practices to maintain and enhance enterprise backend applications.

Technologies: Java, Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Hibernate, REST APIs, Microservices, DB2, SQL, Git

Novac Technology Solutions – Backend Development Intern Aug 2023 – Dec 2023

Chennai, India

- Improved API response times by 40% through SQL query optimization, caching, and efficient payload handling, contributing to a 25% increase in user satisfaction.
- Replaced Webpack with ESBuild, reducing bundling time by 80% and accelerating CI/CD pipeline execution.
- Developed and maintained backend functionality using Express.js and MySQL, while working with Git and Nginx.

Technologies: Express.js, MySQL, jQuery, Git, Nginx, ESBuild

PROJECTS

Resource Booking System Java 21, Spring Boot 4, MySQL, Docker, Gradle

[GitHub]

- Built a backend resource-booking platform using Java and Spring Boot with RESTful APIs for managing users, resources, and bookings.
- Implemented JWT-based authentication and role-based access control using Spring Security to secure application endpoints.
- Designed a layered backend architecture using Controllers, Services, Repositories, and domain entities, following clean separation of concerns.
- Implemented database persistence using Spring Data JPA/Hibernate and MySQL, including relationships between users, resources, and bookings.
- Wrote unit tests for service-layer booking validation logic using JUnit and Mockito to improve code reliability.
- Containerized the application using Docker and prepared an AWS-oriented deployment workflow; used Gradle for dependency management and Postman for API testing.

Other Projects

- AI-Based Traffic Signal Control (Python, OpenCV, YOLO, MQTT) – CV prototype detecting real-time vehicle density to adjust signal timing, with MQTT-based emergency override and manual control. [GitHub]
- Smart Campus Dashboard (Next.js, Python, OpenCV) – cut resource allocation time by 20%. [GitHub]
- Post Invest-Pro (Express.js, MySQL, Next.js, Tailwind CSS) – investment platform for 500+ agents/brokers; +30% processing efficiency. [GitHub]

EDUCATION

Vellore Institute of Technology (VIT) – B. Tech, Information Technology Jul 2021 – Jul 2025

Vellore, India | CGPA: 8.76/10

TECHNICAL SKILLS

Backend: Core Java, Java 8/11/17/21, Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Hibernate, RESTful APIs, Microservices

Databases: DB2, MySQL, MongoDB, SQL, Query Optimization

Testing & Tools: JUnit, Mockito, Postman, Git, GitHub

Cloud & DevOps: AWS, Docker, Maven, Gradle, Nginx

Software Engineering: DSA, OOD, Design Patterns, System Design(Caching, Load Balancing, Indexing), Agile

Languages & Frontend: Java, Python, Typescript/Javascript, HTML5, CSS3, Bootstrap, React.js, Next.js, Node.js